

NEAR SIDE *northern hemisphere*

Taurus-Littrow Valley



TYPE Valley

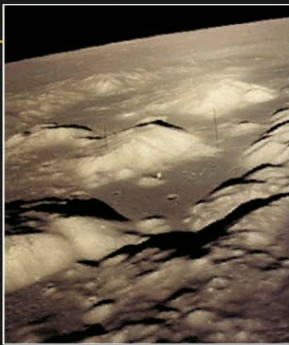
AGE About 3.85 billion years

LENGTH 18.6 miles (30 km)

In December 1972, the last crewed mission to the Moon landed in the dark-floored Taurus-Littrow Valley at the edge of the basalt-filled Mare Serenitatis. The range of geological features was impressive, and the Apollo 17 astronauts found three distinct types of rock in the region. One piece of crushed magnesium olivine was 4.6 billion years old and had crystallized directly from the melted shell of the just-formed Moon. The nearby Serenitatis Crater

HIGHLAND MASSIFS

The flat-based Taurus-Littrow Valley can be seen in the center of this image, nestling between the rugged, blocky mountains known prosaically as the North, South, and East Massifs.



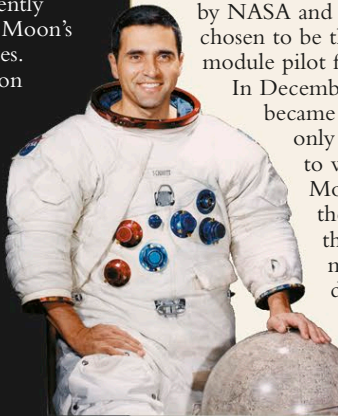
was produced about 3.9 billion years ago, and much of the basaltic rock dates from that time, when the crater was flooded with lava. The third type of rock was found on the top of nearby hills. This was barium-rich granite and had been ejected from one of the surrounding large craters. Most of the material near the landing site was extremely dark and consisted of cinders and ash ejected billions of years ago from nearby volcanic vents and fissures.

like the Alps and Himalayas. These new mountains are subsequently eroded by rain and ice. The Moon's crust is not broken into plates. Nothing moves. All the Moon mountains are produced by impacts, and the mountains around the Taurus-Littrow Valley are the remains of old crater walls. Part of the valley floor just to the north of South

HARRISON SCHMITT

Harrison "Jack" Schmitt (b. 1935) was born in New Mexico. He studied geology at Caltech and Harvard University. While working for the US Geological Survey, he joined a team instructing astronauts in the art of field geology. In June 1965, Schmitt was selected as a scientist-astronaut by NASA and was later chosen to be the lunar module pilot for Apollo 17.

In December 1972, he became the first and only geologist to walk on the Moon. One of the highlights of the Apollo 17 mission was his discovery of orange glass within the lunar rock.



SHORTY CRATER

Harrison Schmitt stands by the Lunar Rover, parked to the left of the 356-ft (110-m) wide Shorty Crater. Behind Schmitt, 4 miles (6 km) away, is Family Mountain, one of the Taurus-Littrow range named by the astronauts. Near the rover, patches of orange soil can be seen.